Power Question 2010: Power of Circular Subdivisions

A king strapped for cash is forced to sell off his kingdom $U = \{(x, y) : x^2 + y^2 \leq 1\}$. He sells the two circular plots C and C' centered at $\left(\pm\frac{1}{2}, 0\right)$ with radius $\frac{1}{2}$. The retained parts of the kingdom form two regions, each bordered by three arcs of circles; in what follows, we will call such regions *curvilinear triangles,* or *c-triangles* (c$\triangle$) for short.

This sad day marks day 0 of a new fiscal era. Unfortunately, these drastic measures are not enough, and so each day thereafter, court geometers mark off the largest possible circle contained in each c-triangle in the remaining property. This circle is tangent to all three arcs of the c-triangle, and will be referred to as the *incircle* of the c-triangle. At the end of the day, all incircles demarcated that day are sold off, and the following day, the remaining c-triangles are partitioned in the same manner.

1. Without using Descartes' Circle Formula (see below):

   a. Show that the circles marked off and sold on day 1 are centered at $\left(0, \pm\frac{2}{3}\right)$ with radius $\frac{1}{3}$. [2]

   b. Find the combined area of the six remaining curvilinear territories. [2]

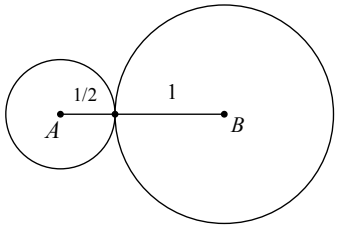
On day 2, the plots bounded by the incircles of the six remaining curvilinear territories are sold.

2a. Determine the number of curvilinear territories remaining at the *end* of day 3. [2]

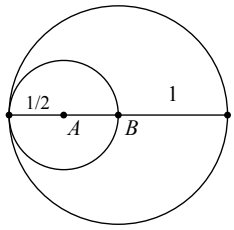
2b. Let X_n be the number of plots sold on day n. Find a formula for X_n in terms of n. [2]

2c. Determine the total number of plots sold up to and including day n. [2]

Some notation: when discussing mutually tangent circles (or arcs), it is convenient to refer to the curvature of a circle rather than its radius. We define *curvature* as follows. Suppose that circle A of radius r_a is externally tangent to circle B of radius r_b. Then the curvatures of the circles are simply the reciprocals of their radii, $\frac{1}{r_a}$ and $\frac{1}{r_b}$. If circle A is internally tangent to circle B, however, as in the right diagram below, the curvature of circle A is still $\frac{1}{r_a}$, while the curvature of circle B is $-\frac{1}{r_b}$, the opposite of the reciprocal of its radius.



Circle A has curvature 2; circle B has curvature 1.



Circle A has curvature 2; circle B has curvature -1.

Using these conventions allows us to express a beautiful theorem of Descartes: when four circles A, B, C, D are pairwise tangent, with respective curvatures a, b, c, d, then

$$(a + b + c + d)^2 = 2\left(a^2 + b^2 + c^2 + d^2\right),$$

where (as before) a is taken to be negative if B, C, D are internally tangent to A, and correspondingly for b, c, or d. This Power Question does not involve the proof of Descartes' Circle Formula, but the formula may be used for all problems below.

3a. Two unit circles and a circle of radius $\frac{2}{3}$ are mutually externally tangent. Compute all possible values of r such that a circle of radius r is tangent to all three circles. [2]

3b. Given three mutually tangent circles with curvatures $a, b, c > 0$, suppose that $(a, b, c, 0)$ does not satisfy Descartes' Circle Formula. Show that there are two distinct values of r such that there is a circle of radius r tangent to the given circles. [3]

3c. Algebraically, it is possible for a quadruple $(a, b, c, 0)$ to satisfy Descartes' Circle Formula, as occurs when $a = b = 1$ and $c = 4$. Find a geometric interpretation for this situation. [2]

4. Let $\phi = \dfrac{1+\sqrt{5}}{2}$, and let $\rho = \phi + \sqrt{\phi}$.

   a. Prove that $\rho^4 = 2\rho^3 + 2\rho^2 + 2\rho - 1$. [2]

   b. Show that four pairwise externally tangent circles with nonequal radii in geometric progression must have common ratio ρ. [2]

As shown in problem 3, given A, B, C, D as above with $s = a + b + c + d$, there is a second circle A' with curvature a' also tangent to B, C, and D. We can describe A and A' as *conjugate circles*.

5. Use Descartes' Circle Formula to show that $a' = 2s - 3a$ and therefore $s' = a' + b + c + d = 3s - 4a$. [4]

In the context of this problem, a *circle configuration* is a quadruple of real numbers (a, b, c, d) representing curvatures of mutually tangent circles A, B, C, D. In other words, a circle configuration is a quadruple (a, b, c, d) of real numbers satisfying Descartes' Circle Formula.

The result in problem 5 allows us to compute the curvatures of the six plots removed on day 2. In this case, $(a, b, c, d) = (-1, 2, 2, 3)$, and $s = 6$. For example, one such plot is tangent to both of the circles C and C' centered at $\left(\pm\frac{1}{2}, 0\right)$, and to one of the circles of radius $\frac{1}{3}$ removed on day 1; it is conjugate to the unit circle (the boundary of the original kingdom U). If this new plot's curvature is a, we can write $(-1, 2, 2, 3) \vdash (a, 2, 2, 3)$, and we say that the first circle configuration *yields* the second.

6a. Use the result of problem 5 to compute the curvatures of all circles removed on day 2, and the corresponding values of s'. [2]

6b. Show that by area, 12% of the kingdom is sold on day 2. [1]

6c. Find the areas of the circles removed on day 3. [2]

6d. Show that the plots sold on day 3 have mean curvature of 23. [1]

7. Prove that the curvature of each circular plot is an integer. [3]

Descartes' Circle Formula can be extended by interpreting the coordinates of points on the plane as complex numbers in the usual way: the point (x, y) represents the complex number $x + yi$. On the complex plane, let z_A, z_B, z_C, z_D be the centers of circles A, B, C, D respectively; as before, a, b, c, d are the curvatures of their respective circles. Then Descartes' Extended Circle Formula states

$$(a \cdot z_A + b \cdot z_B + c \cdot z_C + d \cdot z_D)^2 = 2\left(a^2 z_A^2 + b^2 z_B^2 + c^2 z_C^2 + d^2 z_D^2\right).$$

8a. Suppose that A' is a circle conjugate to A with center $z_{A'}$ and curvature a', and $\hat{s} = a \cdot z_A + b \cdot z_B + c \cdot z_C + d \cdot z_D$. Use Descartes' Extended Circle Formula to show that $a' \cdot z_{A'} = 2\hat{s} - 3a \cdot z_A$ and therefore $a' \cdot z_{A'} + b \cdot z_B + c \cdot z_C + d \cdot z_D = 3\hat{s} - 4a \cdot z_A$. [2]

8b. Prove that the center of each circular plot has coordinates $\left(\frac{u}{c}, \frac{v}{c}\right)$ where u and v are integers, and c is the curvature of the plot. [2]

Given a c-triangle T, let a, b, and c be the curvatures of the three arcs bounding T, with $a \le b \le c$, and let d be the curvature of the incircle of T. Define the *circle configuration associated with* T to be $\mathcal{C}(T) = (a, b, c, d)$. Define the c-triangle T to be *proper* if $c \le d$. For example, circles of curvatures -1, 2, and 3 determine two c-triangles. The incircle of one has curvature 6, so it is proper; the incircle of the other has curvature 2, so it is not proper.

Let P and Q be two c-triangles, with associated configurations $\mathcal{C}(P) = (a, b, c, d)$ and $\mathcal{C}(Q) = (w, x, y, z)$. We say that P *dominates* Q if $a \le w$, $b \le x$, $c \le y$, and $d \le z$. (The term "dominates" refers to the fact that the radii of the arcs defining Q cannot be larger than the radii of the arcs defining P.)

Removing the incircle from T gives three c-triangles, $T^{(1)}$, $T^{(2)}$, $T^{(3)}$, each bounded by the incircle of T and two of the arcs that bound T. These triangles have associated configurations

$$\begin{aligned} \mathcal{C}(T^{(1)}) &= (b, c, d, a'), \\ \mathcal{C}(T^{(2)}) &= (a, c, d, b'), \\ \mathcal{C}(T^{(3)}) &= (a, b, d, c'), \end{aligned}$$

where a', b', and c' are determined by the formula in problem 5.

9. Let P and Q be two proper c-triangles such that P dominates Q. Let $\mathcal{C}(P) = (a, b, c, d)$ and $\mathcal{C}(Q) = (w, x, y, z)$.

   a. Show that $P^{(3)}$ dominates $P^{(2)}$ and that $P^{(2)}$ dominates $P^{(1)}$. [2]

   b. Prove that $P^{(1)}$ dominates $Q^{(1)}$. [2]

   c. Prove that $P^{(3)}$ dominates $Q^{(3)}$. [2]

10a. Prove that the largest plot sold by the king on day n has curvature $n^2 + 2$. [3]

10b. If $\rho = \phi + \sqrt{\phi}$, as in problem 4, prove that the curvature of the smallest plot sold by the king on day n does not exceed $2\rho^n$. [3]